Sequence Diagrams: RAG-Diary

> The message sequence between system components for the important use cases, showing both the success path and the failure path, so the data flow is visible at message level.

> Thai version: SEQUENCE-DIAGRAMS.md

1. Document control

Item	Detail
Document	Sequence Diagrams
Version	1.0
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
Status	Reviewed, round 1 (25 October 2026)
Related documents	USE-CASES, C4-ARCHITECTURE, API-CONTRACT

2. Purpose and scope

This document answers one question: when a user does something, how does the request travel from the screen to the server, the database and the external providers, and what happens when it fails?

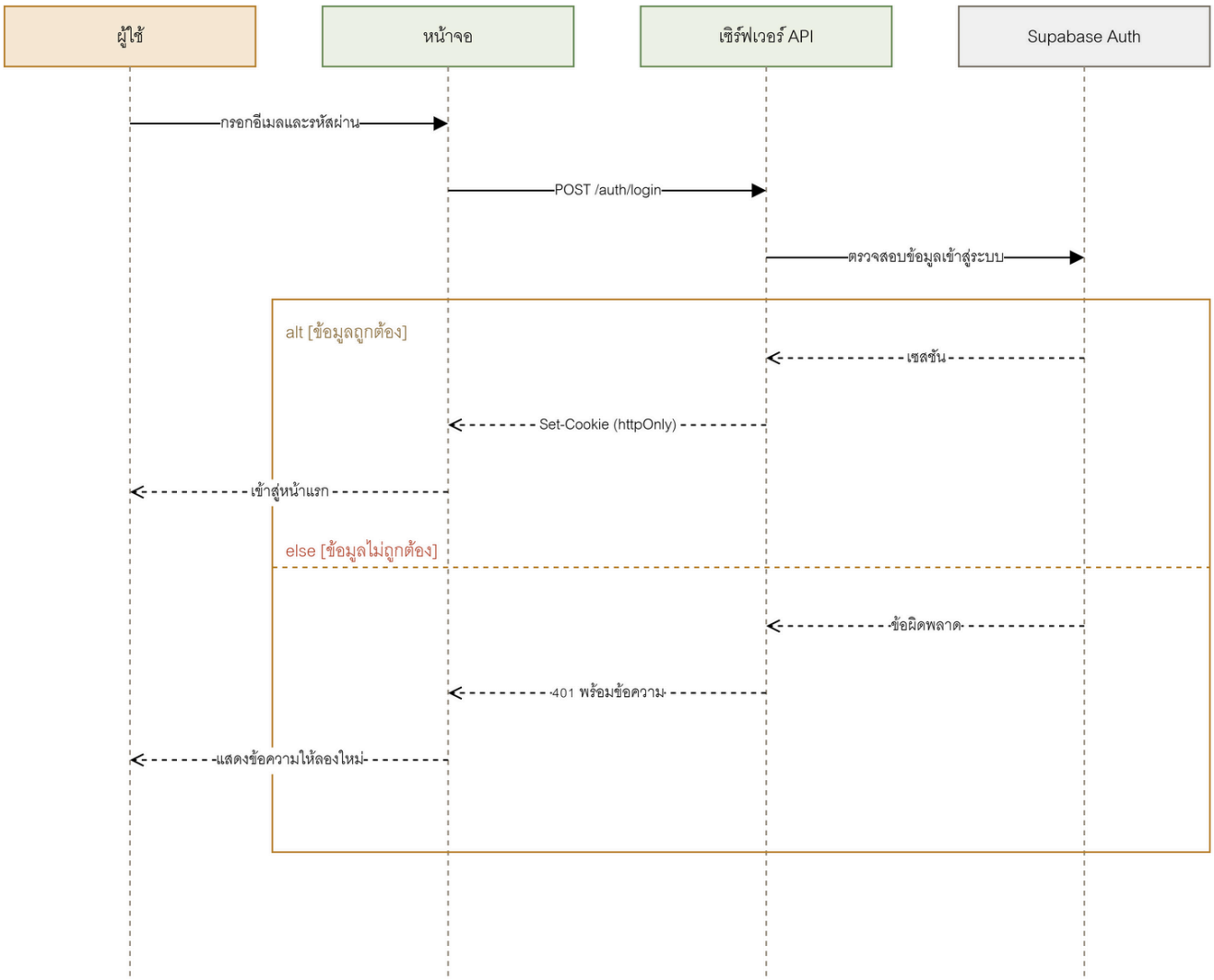
Scope: six message sequences: login, writing an entry with its index, semantic search, question answering, deleting all data, and renewing the session. Out of scope: role-level process models (in PROCESS-FLOWS).

3. Summary table

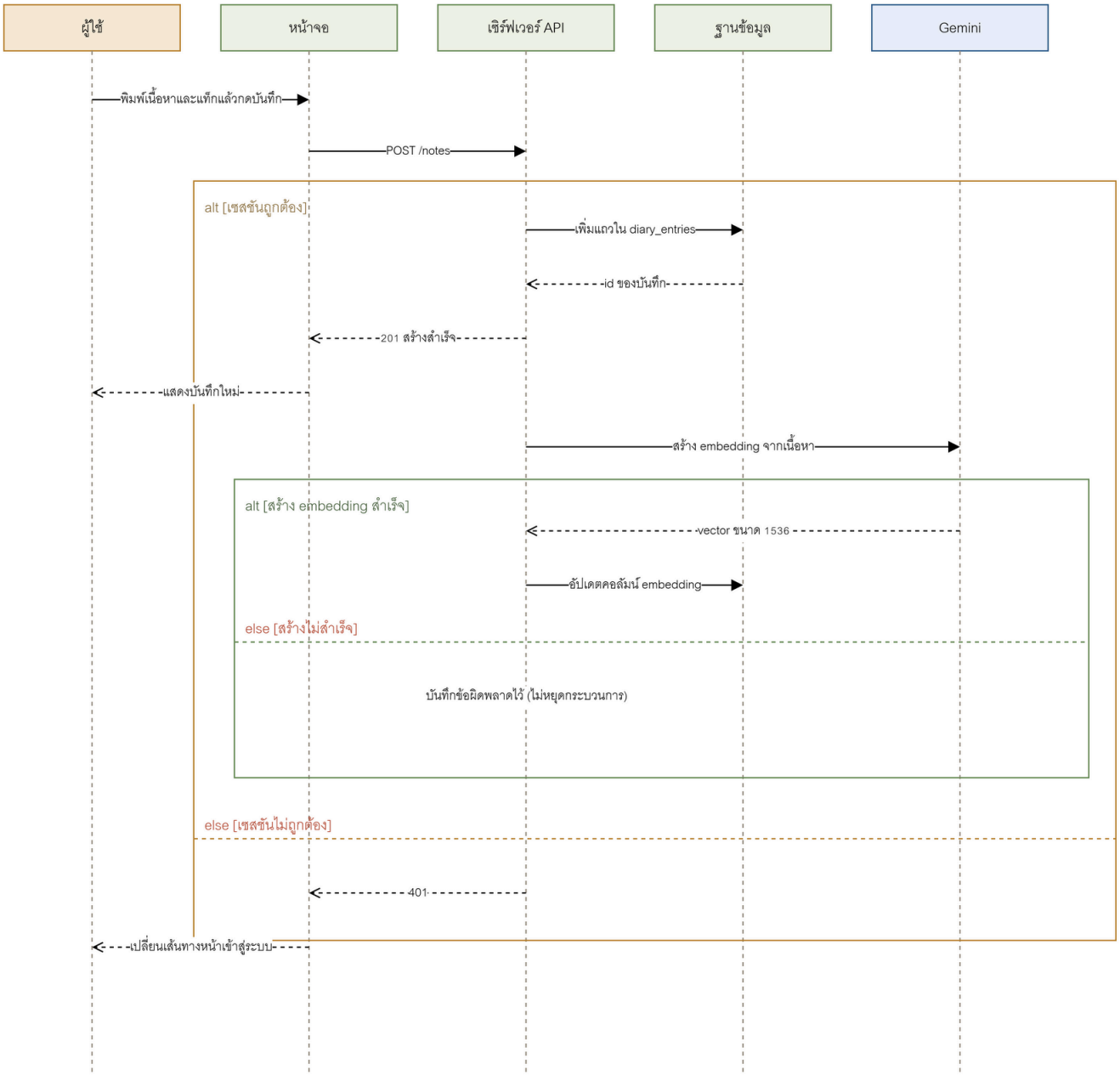
ID	Use case	Participants	Success path	Failure path
SEQ-01	UC-01 Log in	User, screen, API, Supabase Auth	Reaches the home page	Shows a message and offers a retry
SEQ-02	UC-02 Write an entry and build the	User, screen, API, DB, Gemini	The entry is stored with its index	Notifies the user and offers a retry

ID	Use case	Participants	Success path	Failure path
	index			
SEQ-03	UC-04 Semantic search	User, screen, API, Gemini, DB	Search results are shown	Reports that nothing was found
SEQ-04	UC-04 Chat with the entries	User, screen, API, Gemini, DB	An answer is shown with references	Asks the user to retry when a service fails
SEQ-05	UC-06 Delete all data	User, screen, API, DB	Everything is deleted and the cache is cleared	Cancelling leaves the data untouched
SEQ-06	Renew the session	Screen, API, Supabase Auth	A new session is issued and work continues	The session is cleared and the user returns to login

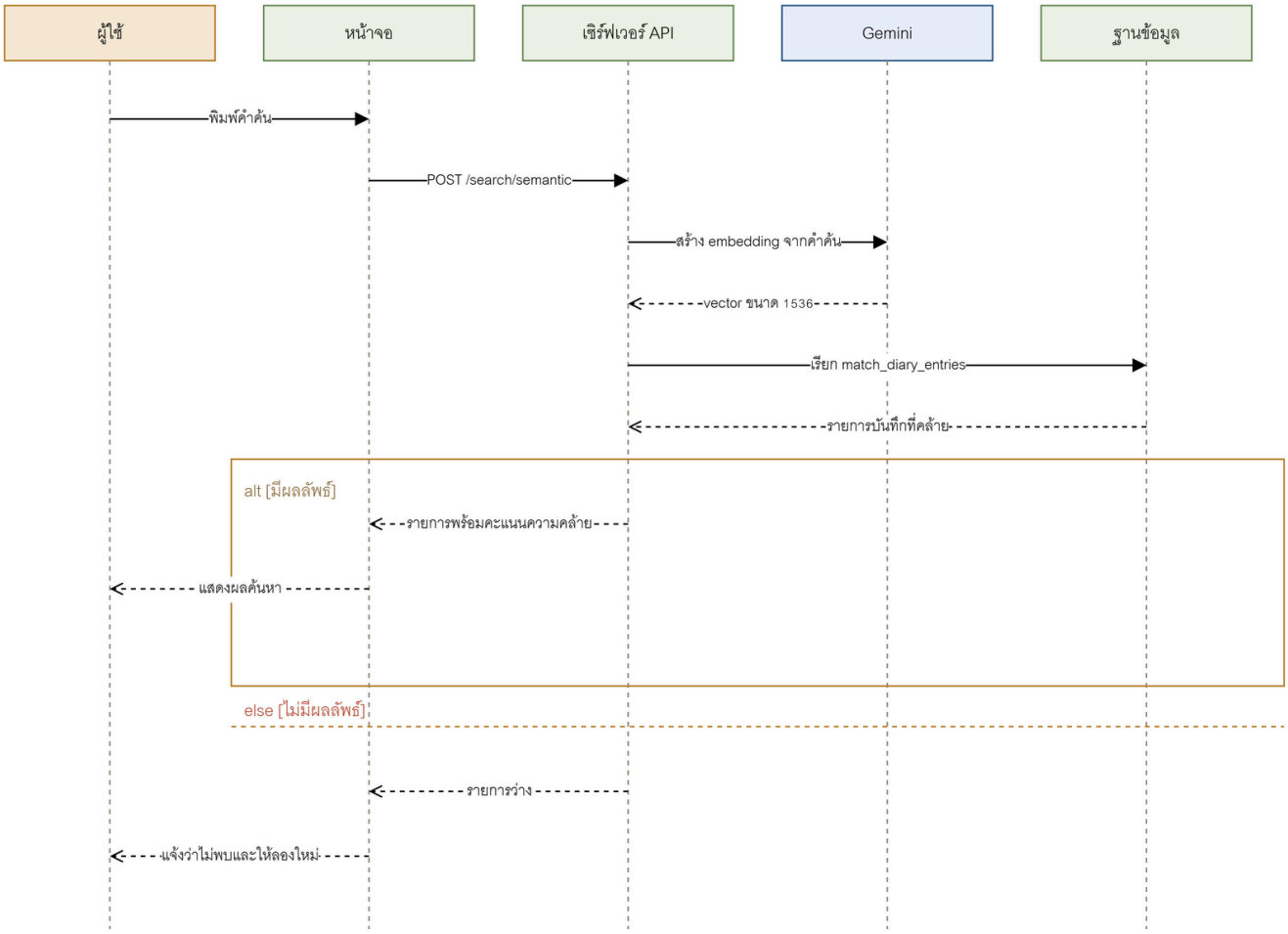
4. SEQ-01 Log in



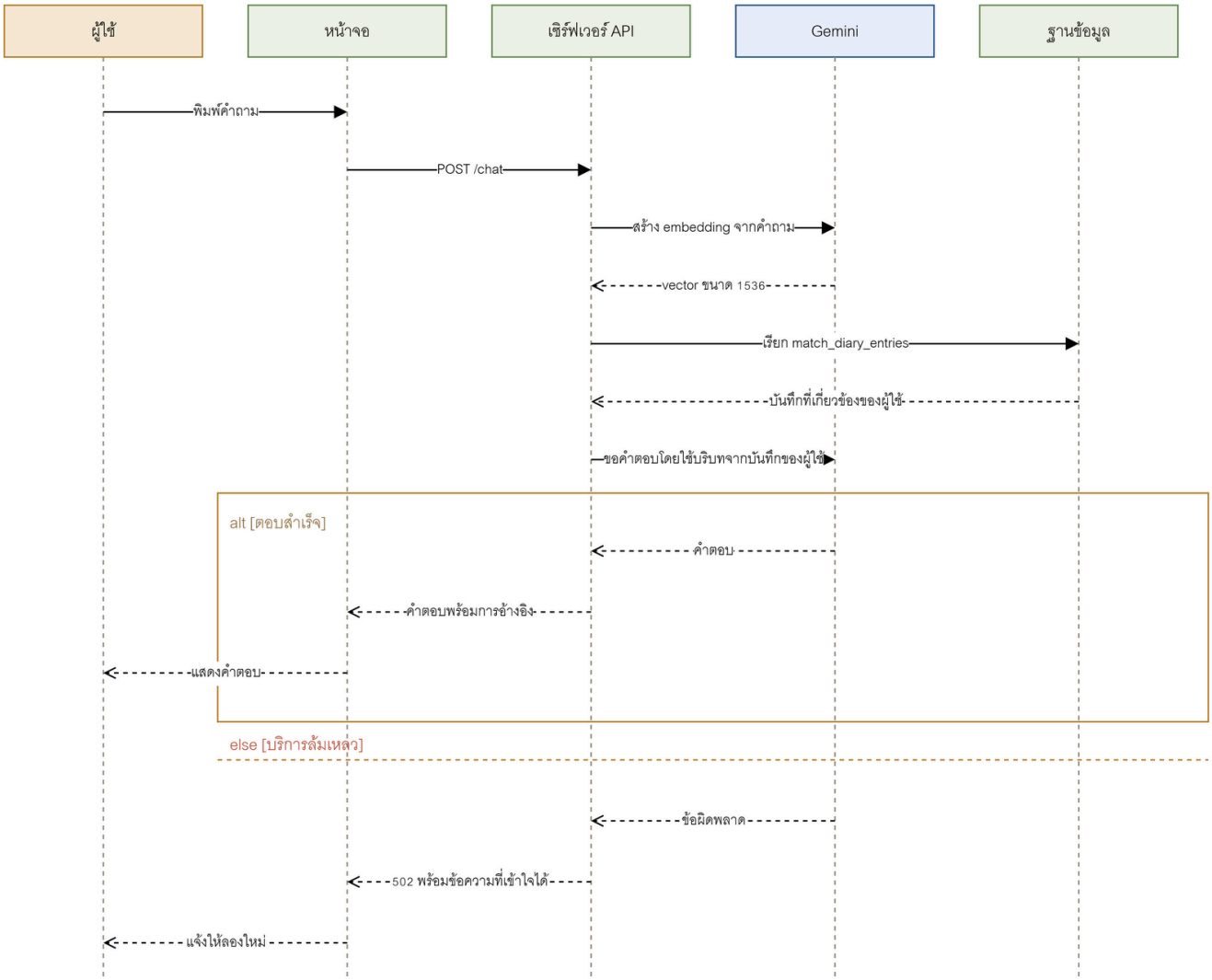
5. SEQ-02 Write an entry and build the index



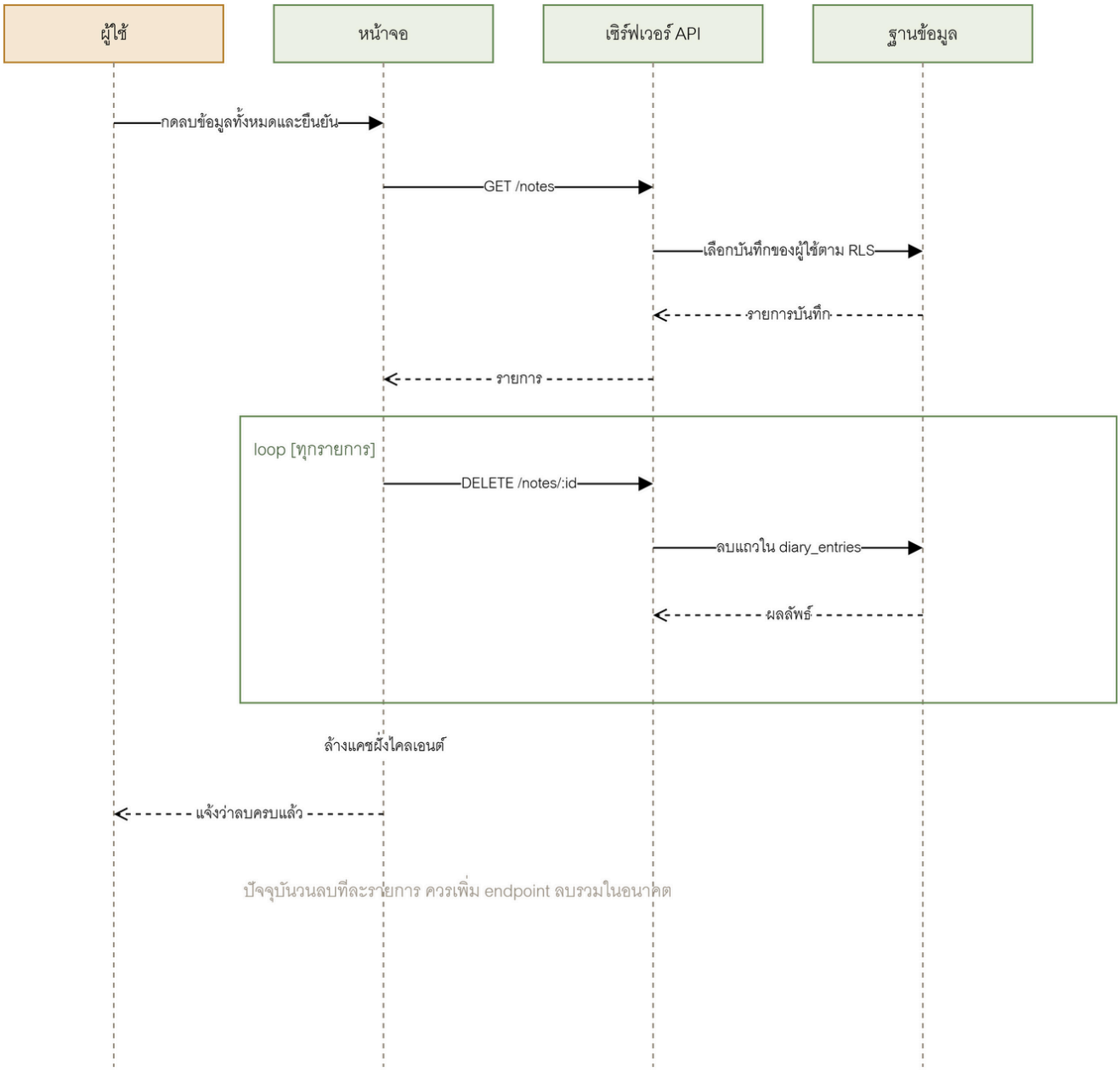
6. SEQ-03 Semantic search



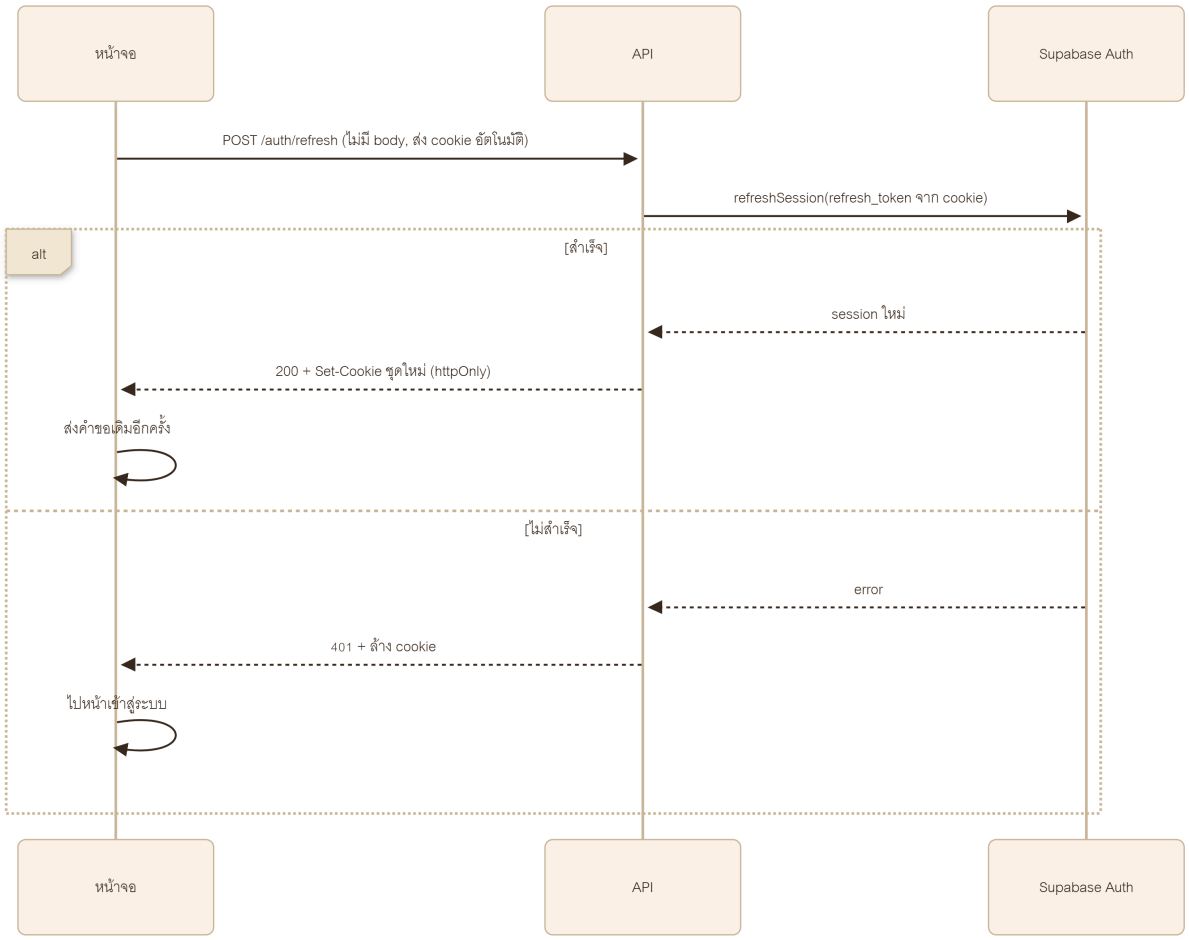
7. SEQ-04 Chat with the entries



8. SEQ-05 Delete all data



9. SEQ-06 Renew the session



10. Notes

- The diagrams show only the interactions needed to understand the flow; log writing at every point is omitted.
- The order in SEQ-04 reflects the RAG pattern: build the index, retrieve the context, then generate the answer.
- Delete-all in SEQ-05 uses DELETE /data, added later, which closes the gap once recorded in the RTM.

11. References

Document	Relationship
USE-CASES	The use cases each SEQ comes from
C4-ARCHITECTURE	The components shown in the diagrams
API-CONTRACT	The contract of the endpoints being called
DATA-MODEL	The tables and functions being called